

DIFFERENTIAL CLINICAL PREDICTION OF INATTENTION AND HYPERACTIVITY IN CHILDREN USING TASK-BASED DIGITAL PHENOTYPING

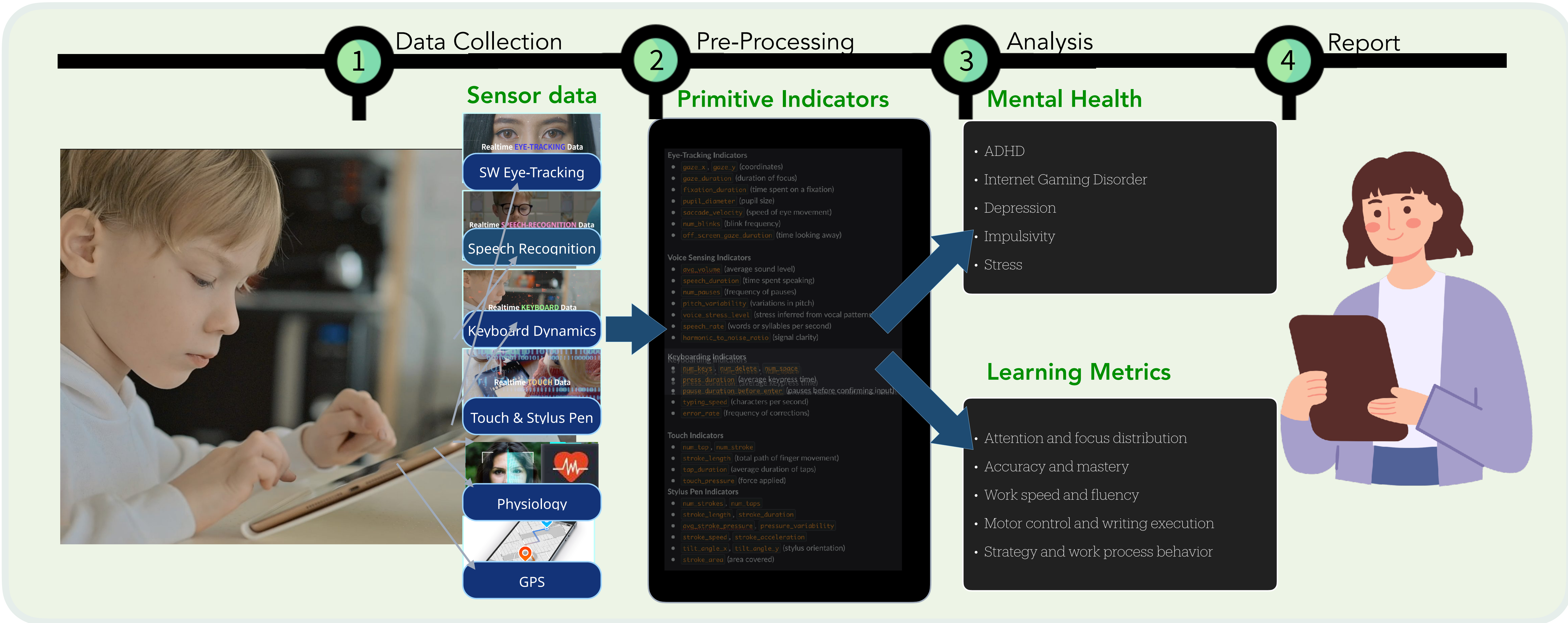
Kukju Kwon, MD, Ph.D.^{1,2,3}; Mahsa Aghazadeh, MS⁴; Su A. Kwon, Ph.D.¹; Seonghyeon Park, ME⁴; Kwangsu Cho, Ph.D.⁴

¹ Department of Psychiatry, Chungnam National University Hospital, Daejeon, Republic of Korea,

² Department of Psychiatry, Chungnam National University School of Medicine, Daejeon, Republic of Korea

³ Department of Psychiatry, University of Pittsburgh School of Medicine, Pittsburgh, PA, USA

⁴ 3R Innovation Research Center, Seoul, Republic of Korea



INTRODUCTION

Attention-deficit/hyperactivity disorder in children is clinically heterogeneous, and its core domains of inattention and hyperactivity differ in behavioral expression, functional significance, and detectability during structured tasks. Digital phenotyping may therefore be more clinically informative when these domains are modeled separately rather than collapsed into a single ADHD outcome.

OBJECTIVE

This study examined whether task-based digital phenotypes derived from an integrated computerized assessment platform could predict clinician-rated inattention and hyperactivity in children, and whether separate symptom-domain modeling would be more informative than combined ADHD prediction.

METHODS

103 children completed a standardized digital neurocognitive battery within the Dr. Simon-based Psychological Decision Support Service platform in a clinical setting. The battery included tasks assessing visual and auditory attention, inhibitory control, interference processing, and working memory. The platform automatically extracted behavioral markers including accuracy, omission and commission errors, response speed, response variability, and temporal indicators of instability and performance drift. Outcomes were referenced against clinician-rated symptoms based on K-SADS-COMP.

RESULTS

Separate symptom-domain models showed domain-specific performance. For inattention, the best classifier achieved an accuracy of 0.704, ROC-AUC of 0.722, and F2 of 0.660, while severity prediction was limited ($r = 0.326$, negative R^2). For hyperactivity, the best classifier achieved an accuracy of 0.784, ROC-AUC of 0.755, and F2 of 0.647, with stronger severity prediction ($r = 0.539$, $R^2 = 0.209$). Compared with the combined ADHD model (accuracy 0.680, ROC-AUC 0.720, F2 0.516), separate symptom-domain modeling provided more robust and clinically interpretable prediction.

CONCLUSIONS

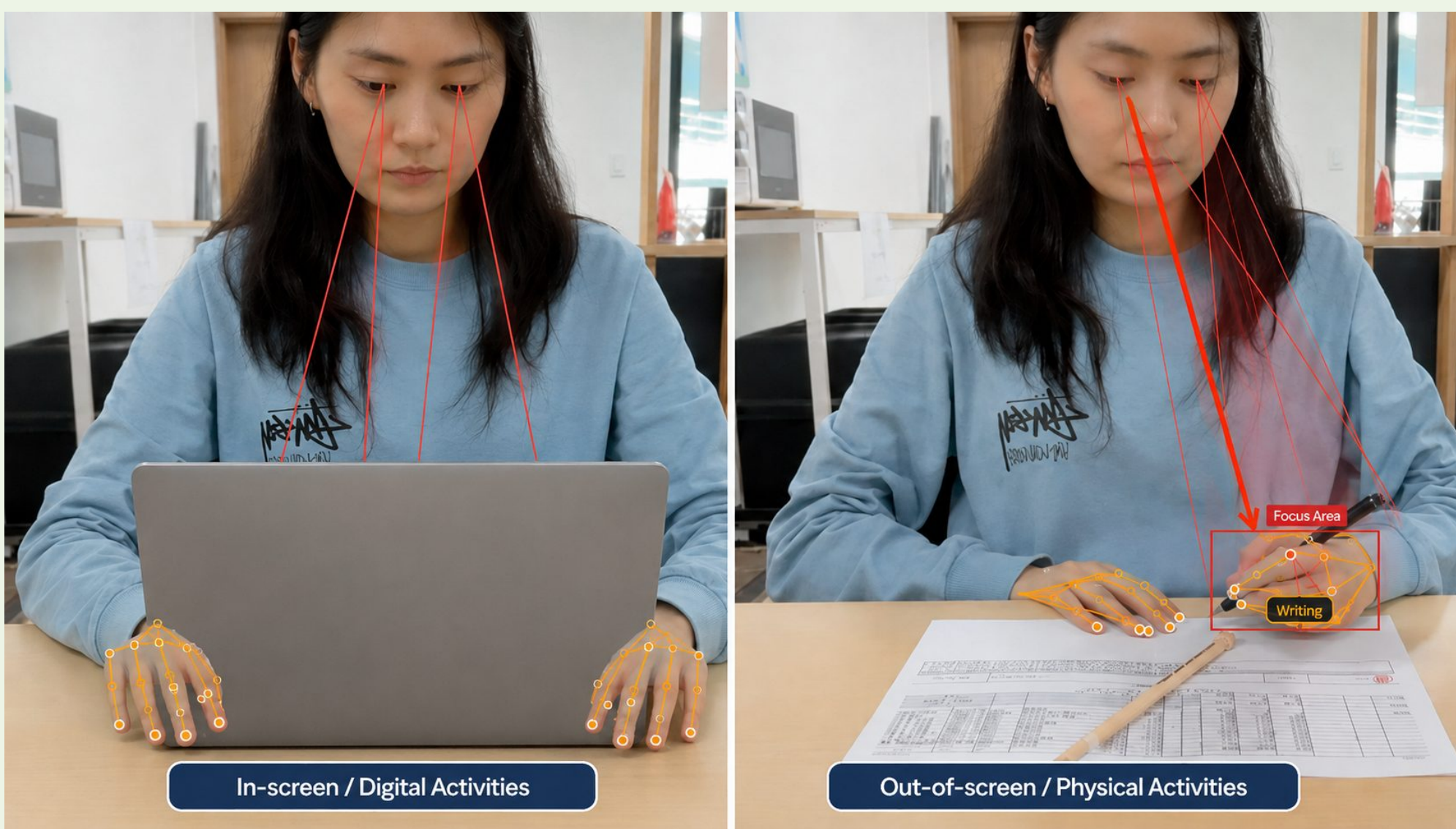
Task-based digital phenotyping may be more clinically useful when inattention and hyperactivity are modeled separately. Integrated digital platforms may therefore serve as adjunctive tools for pediatric ADHD triage and longitudinal monitoring, rather than as stand-alone diagnostic instruments.

INTERESTED IN COLLABORATION?

Contact Kwangsu Cho at kc@focuspang.com.

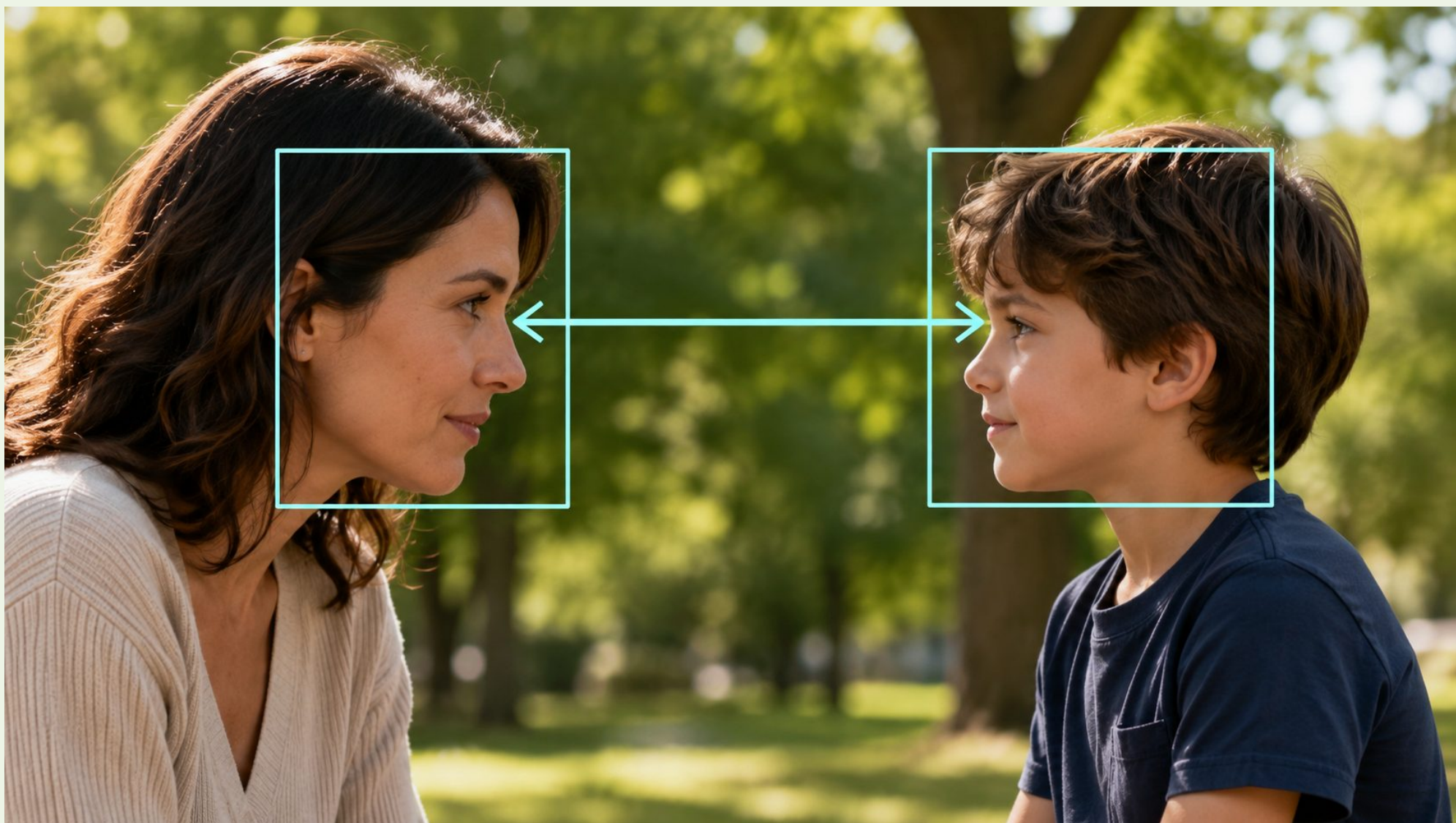
ADHD Research During Digital and Paper-Based Activities in Everyday Settings

Detect inattention, hyperactivity, impulsivity, and task persistence during digital and paper-based activities.



Joint Attention for Autism Spectrum Disorder

Study how children share attention with others through gaze, pointing, and gestures during everyday interactions.



CHUNGNAM
NATIONAL
UNIVERSITY
HOSPITAL



UPMC
University of Pittsburgh
Medical Center

